

LOSO Experiment A1

Full sensors (flex+ypr+accel, both hands), INCLUDING Double_Nothing class, 11-subject LOSO

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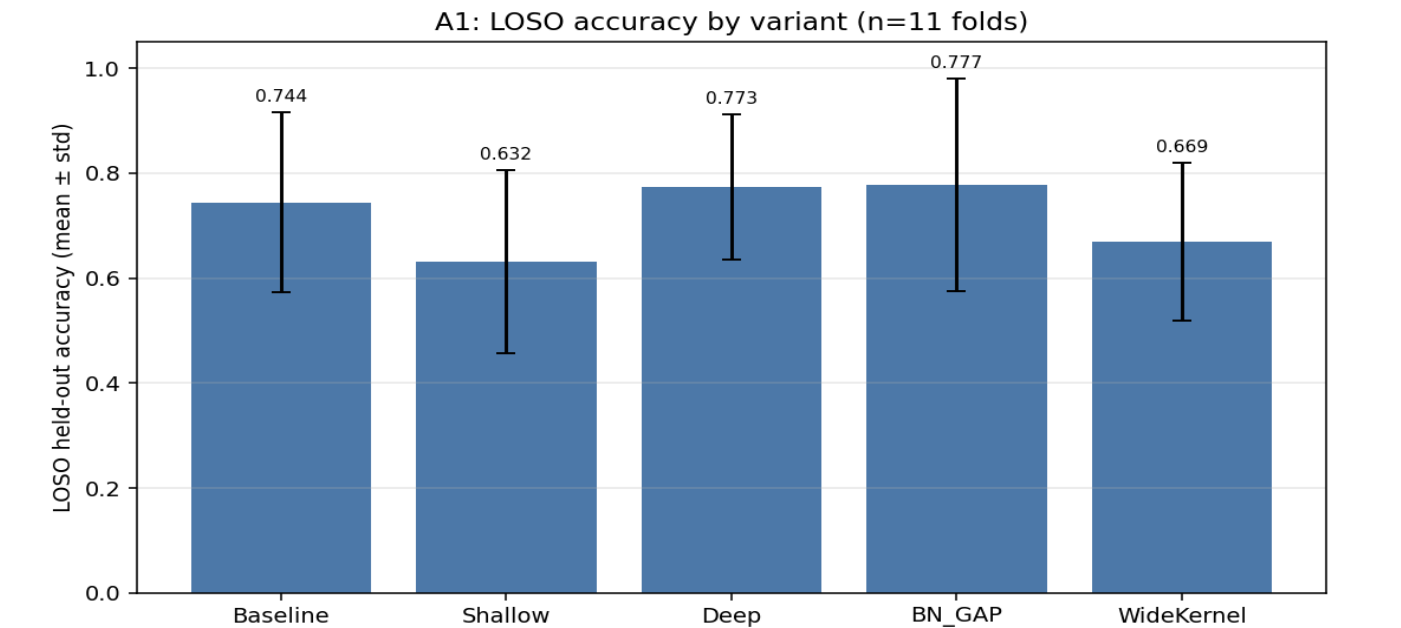
Experiment configuration

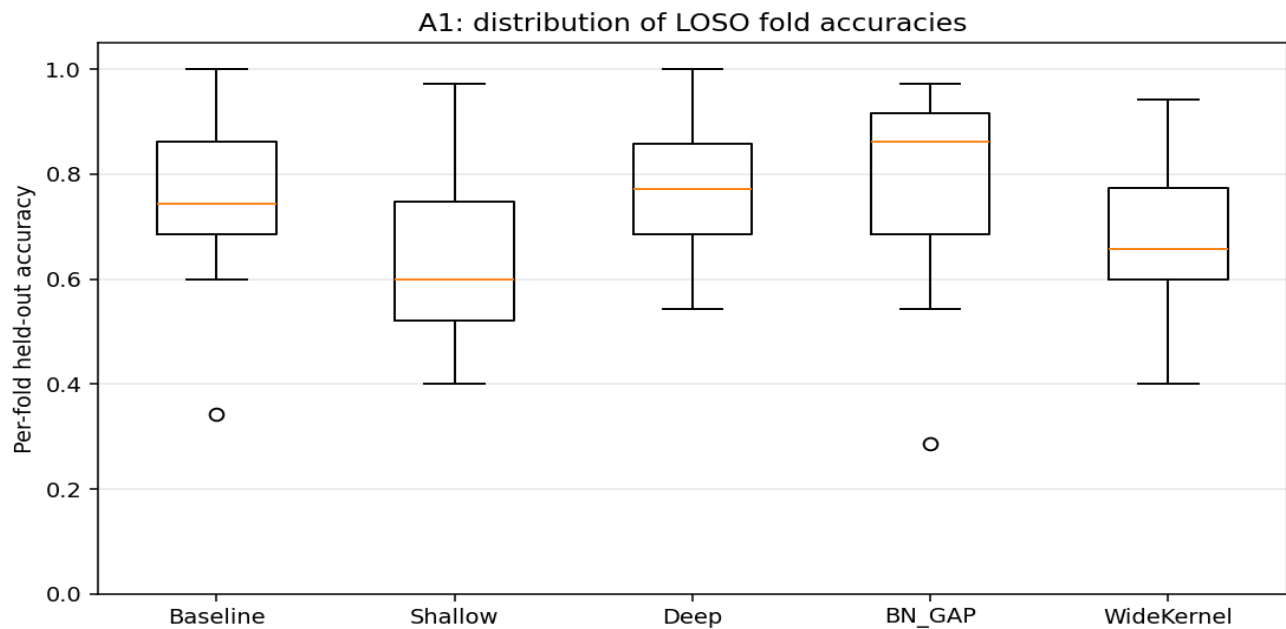
Class set	7 classes including Double_Nothing
Classes	Double_Lower, Double_Nothing, Double_Pistol_Recoil, Double_Raise, Double_Wiggle, Double_cmere, Drum_Roll
Sensor modalities	flex, YPR, accel
Hands	left, right
Sensor channels	176
Sequence length	90 steps (resampled)
Filter	Butterworth 10.0Hz order 4
Normalisation	minmax
Augmentation	amplitude_scale + baseline_offset, 3 copies
Inner CV folds	3
Epochs · batch size	8 · 16
LOSO subjects	11: Alan, Alex, Anghad, Bella, Daniel, Harry, Marsh, Marcus, Raquel, StephenV2, Tash

LOSO summary across all subjects

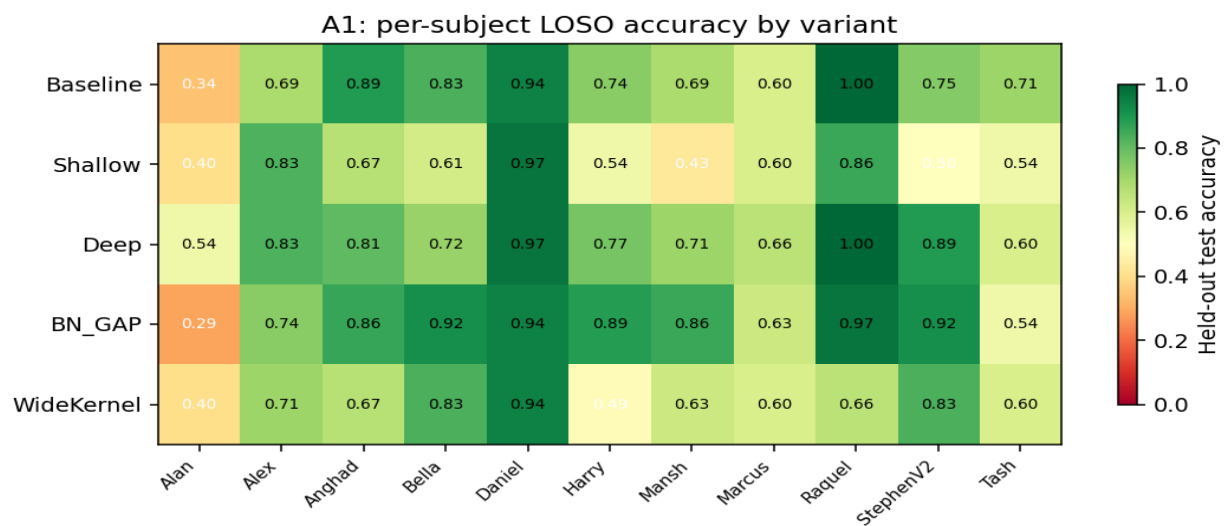
Variant	Params	LOSO mean	LOSO std	Min fold	Max fold	Inner CV mean (avg across folds)
Baseline	113,255	0.7442	0.1715	0.3429	1.0000	0.8583
Shallow	72,487	0.6317	0.1747	0.4000	0.9714	0.7359
Deep	204,007	0.7729	0.1384	0.5429	1.0000	0.8977
BN_GAP	57,319	0.7774	0.2020	0.2857	0.9714	0.9531
WideKernel	131,687	0.6693	0.1502	0.4000	0.9429	0.7920

Best variant by LOSO mean: **BN_GAP**





Per-subject LOSO accuracy heatmap



Per-fold detail (held-out subject = each row)

Held-out subj	Train trials	Baseline	Shallow	Deep	BN_GAP	WideKernel
Alan	353	0.343	0.400	0.543	0.286	0.400
Alex	353	0.686	0.829	0.829	0.743	0.714
Anghad	352	0.889	0.667	0.806	0.861	0.667
Bella	352	0.833	0.611	0.722	0.917	0.833
Daniel	353	0.943	0.971	0.971	0.943	0.943
Harry	353	0.743	0.543	0.771	0.886	0.486
Mansh	353	0.686	0.429	0.714	0.857	0.629
Marcus	353	0.600	0.600	0.657	0.629	0.600
Raquel	353	1.000	0.857	1.000	0.971	0.657
StephenV2	352	0.750	0.500	0.889	0.917	0.833
Tash	353	0.714	0.543	0.600	0.543	0.600

Variant: Baseline

Params: 113,255

LOSO mean accuracy: 0.7442 ± 0.1715 (min 0.3429, max 1.0000)

Per-fold per-class held-out accuracy

Held-out	Double_Lower	Double_Nothing	Double_Pistol_Recoil	Double_Raise	Double_Wiggle	Double_cmere	Drum_Roll	Overall
Alan	4/5	2/5	0/5	1/5	5/5	0/5	0/5	0.343
Alex	0/5	5/5	4/5	5/5	0/5	5/5	5/5	0.686
Anghad	5/5	1/5	5/5	5/5	5/5	6/6	5/5	0.889
Bella	5/5	0/6	5/5	5/5	5/5	5/5	5/5	0.833
Daniel	5/5	3/5	5/5	5/5	5/5	5/5	5/5	0.943
Harry	5/5	4/5	3/5	3/5	5/5	4/5	2/5	0.743
Mansh	5/5	0/5	2/5	5/5	5/5	5/5	2/5	0.686
Marcus	5/5	0/5	4/5	5/5	4/5	3/5	0/5	0.600
Raquel	5/5	5/5	5/5	5/5	5/5	5/5	5/5	1.000
StephenV2	5/5	0/6	4/5	5/5	3/5	5/5	5/5	0.750
Tash	5/5	3/5	5/5	5/5	4/5	2/5	1/5	0.714
MEAN	0.891	0.418	0.764	0.891	0.836	0.800	0.636	0.744

Variant: Shallow

Params: 72,487

LOSO mean accuracy: 0.6317 ± 0.1747 (min 0.4000, max 0.9714)

Per-fold per-class held-out accuracy

Held-out	Double_Lower	Double_Nothing	Double_Pistol_Re coil	Double_Raise	Double_Wiggle	Double_cmere	Drum_Roll	Overall
Alan	1/5	3/5	0/5	0/5	5/5	5/5	0/5	0.400
Alex	2/5	5/5	3/5	5/5	4/5	5/5	5/5	0.829
Anghad	5/5	1/5	5/5	5/5	0/5	6/6	2/5	0.667
Bella	5/5	0/6	5/5	3/5	0/5	5/5	4/5	0.611
Daniel	5/5	4/5	5/5	5/5	5/5	5/5	5/5	0.971
Harry	4/5	1/5	5/5	0/5	4/5	0/5	5/5	0.543
Mansh	5/5	0/5	0/5	5/5	0/5	5/5	0/5	0.429
Marcus	5/5	0/5	4/5	5/5	3/5	4/5	0/5	0.600
Raquel	5/5	5/5	5/5	5/5	5/5	5/5	0/5	0.857
StephenV2	0/5	2/6	5/5	3/5	0/5	3/5	5/5	0.500
Tash	5/5	2/5	5/5	5/5	0/5	2/5	0/5	0.543
MEAN	0.764	0.412	0.764	0.745	0.473	0.800	0.473	0.632

Variant: Deep

Params: 204,007

LOSO mean accuracy: 0.7729 ± 0.1384 (min 0.5429, max 1.0000)

Per-fold per-class held-out accuracy

Held-out	Double_Lower	Double_Nothing	Double_Pistol_Recoil	Double_Raise	Double_Wiggle	Double_cmere	Drum_Roll	Overall
Alan	5/5	0/5	0/5	5/5	5/5	4/5	0/5	0.543
Alex	5/5	5/5	0/5	5/5	5/5	5/5	4/5	0.829
Anghad	5/5	2/5	4/5	5/5	5/5	6/6	2/5	0.806
Bella	5/5	1/6	5/5	1/5	5/5	5/5	4/5	0.722
Daniel	5/5	4/5	5/5	5/5	5/5	5/5	5/5	0.971
Harry	5/5	2/5	5/5	5/5	5/5	0/5	5/5	0.771
Mansh	5/5	0/5	0/5	5/5	5/5	5/5	5/5	0.714
Marcus	5/5	0/5	2/5	5/5	5/5	5/5	1/5	0.657
Raquel	5/5	5/5	5/5	5/5	5/5	5/5	5/5	1.000
StephenV2	5/5	5/6	5/5	2/5	5/5	5/5	5/5	0.889
Tash	4/5	0/5	5/5	5/5	4/5	2/5	1/5	0.600
MEAN	0.982	0.418	0.655	0.873	0.982	0.836	0.673	0.773

Variant: BN_GAP

Params: 57,319

LOSO mean accuracy: 0.7774 ± 0.2020 (min 0.2857, max 0.9714)

Per-fold per-class held-out accuracy

Held-out	Double_Lower	Double_Nothing	Double_Pistol_Re coil	Double_Raise	Double_Wiggle	Double_cmere	Drum_Roll	Overall
Alan	0/5	5/5	0/5	1/5	0/5	4/5	0/5	0.286
Alex	5/5	5/5	1/5	4/5	1/5	5/5	5/5	0.743
Anghad	5/5	4/5	5/5	5/5	4/5	6/6	2/5	0.861
Bella	5/5	4/6	5/5	5/5	4/5	5/5	5/5	0.917
Daniel	5/5	3/5	5/5	5/5	5/5	5/5	5/5	0.943
Harry	5/5	3/5	5/5	5/5	5/5	4/5	4/5	0.886
Mansh	5/5	0/5	5/5	5/5	5/5	5/5	5/5	0.857
Marcus	5/5	5/5	4/5	5/5	0/5	3/5	0/5	0.629
Raquel	5/5	4/5	5/5	5/5	5/5	5/5	5/5	0.971
StephenV2	5/5	3/6	5/5	5/5	5/5	5/5	5/5	0.917
Tash	1/5	2/5	4/5	5/5	5/5	2/5	0/5	0.543
MEAN	0.836	0.670	0.800	0.909	0.709	0.873	0.655	0.777

Variant: WideKernel

Params: 131,687

LOSO mean accuracy: 0.6693 ± 0.1502 (min 0.4000, max 0.9429)

Per-fold per-class held-out accuracy

Held-out	Double_Lower	Double_Nothing	Double_Pistol_Recoil	Double_Raise	Double_Wiggle	Double_cmere	Drum_Roll	Overall
Alan	5/5	4/5	0/5	5/5	0/5	0/5	0/5	0.400
Alex	5/5	5/5	0/5	5/5	0/5	5/5	5/5	0.714
Anghad	5/5	2/5	5/5	5/5	0/5	6/6	1/5	0.667
Bella	5/5	0/6	5/5	5/5	5/5	5/5	5/5	0.833
Daniel	5/5	4/5	5/5	5/5	5/5	4/5	5/5	0.943
Harry	5/5	2/5	0/5	0/5	5/5	4/5	1/5	0.486
Mansh	5/5	0/5	1/5	5/5	5/5	5/5	1/5	0.629
Marcus	5/5	1/5	3/5	5/5	2/5	5/5	0/5	0.600
Raquel	5/5	5/5	3/5	5/5	0/5	5/5	0/5	0.657
StephenV2	5/5	0/6	5/5	5/5	5/5	5/5	5/5	0.833
Tash	3/5	3/5	5/5	5/5	4/5	1/5	0/5	0.600
MEAN	0.964	0.473	0.582	0.909	0.564	0.800	0.418	0.669